

? How Does An MFD Work?

Most MFD units comprise a printer unit, a scanner unit, a fax unit and a copier. Almost all the MFDs will have these four functions, some may include a memory card reader. It is, however, not possible to use any two functions at the same time—except for high-end MFDs that support multitasking. The dual functions that a high-end MFD can perform are either fax and print or print and scan. It is quite difficult to have any other combinations.

Let's try and understand each function of printing, scanning, copying and faxing and try to understand how an MFD unit works.

Inkjet Printers

An inkjet printer generally has two cartridges—one colour and another black. These cartridges have slots through which electricity is supplied to the head of the cartridge. The head receives current from both left and right side, which charges it. Once charged the ink within the cartridge is thrown out in the form of a drop and may either burst or simply be deposited on the paper. The lesser the picolitre of

the drop the finer is the quality of printout.

Laserjet Printers

Unlike the inkjet, laserjets do not have cartridges but drums. These drums consist of a toner and a laser point. Once data is received by the PCB, it is transferred via the laser tip and written onto the drum. The drum and toner are both charged oppositely. Now when a paper passes between the drum and the toner they are attracted to each other because of this negative charge and the characters written on the drum is transferred from the drum to the paper. Next, the developer helps the characters to be adsorbed on the paper and the final result is obtained in the form of a printout.

Scanners

Initially whatever was scanned had to be transferred to the computer as a soft copy and then a print taken. Now, however with MFDs a direct printout can be obtained. Flatbed scanners have a light running beneath the transparent surface on which the material to be scanned is

kept. The cover is also generally white in colour so that it reflects the light back. The characters printed on the page have light falling on it from beneath the scanner bed. When this light hits the character an image of it is reproduced on the paper in the tray and a hard copy of the scanned material is obtained. So now with MFDs the trouble of making a soft copy and then taking a print out is eliminated.

Faxing

Faxing involves transmission of data over telephone lines. This involves conversion of analog signals to digital and vice-versa. Rather than the output being in the form of voice it is in the form of data in a fax.

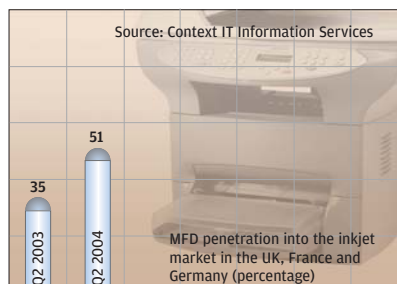
Thus, when you ask an MFD to print you know what happens. Most corporates are not really bothered about the cost difference between an independent unit and a MFD. One of the major constraints in most MFDs is that they can perform only single tasks at a time and the high end MFD's price runs into lakhs of rupee which works as a deterrent.

HP 4410 owner John Fernandes, a trader in chemicals, says "I print nearly 30-40 invoices a week."

Niraj Garg from Cyberpark India, New Delhi, owns an HP 5510 and Laser P380. An IT trader specialising in corporate and retail businesses, Garg is a heavy-duty MFD user. The 5510 is shared by him and two more users whereas the Laser P380 has about 10-11 users configured.

He prints 200 copies per month on the 5510 and 2000 cpm on the Laser P380. "Both products are very good and I do not have any complaints. Yes, I would like to see the prices of these fall though," says Garg.

Rajiv Soni, an LIC agent uses HP 4110, "Computers and technology related to it are my hobby. I use my MFD for sending and receiving fax to and from clients. I would most certainly like to purchase any new and better products coming in the market. I also do a lot of scanning, so with my current system I cannot scan thick books. So that's a problem, would like to see some improvement there", says Soni.



Coming Soon...

A major hindrance in MFD penetration has been customer awareness. Customers in India are yet to realise the strong value proposition of MFDs. Though vendors are trying their best to increase awareness the message has been slow in percolation. The inertia of a hesitancy in adopting a new concept has been some hindrance. Inkjet is a technology will die in the coming few years.

Laser is the technology of the future with a number of inherent customer benefits like quality, lower cost, etc. Globally MFDs are bringing in a huge change in terms of the way enterprise business is run.

MFDs, the way we see them today, have only emerged in the last few of years and prices have fallen drastically, as the category itself gets bigger. Initially the challenge was to create awareness of the multiple uses this product would likely have towards cutting the total cost of ownership and in providing value for money. A few years ago, their costs may have been prohibitive, but today entry level MFDs are available for as low as Rs 15,000.

Earlier there were limited products on offer, and perhaps a handful of players catering to all the consumers be it the end user, SoHo, SMBs or large corporates. Today the entry of MFDs as the new offering has created a buzz in the corporate market, and players are trying to position the products aggressively. One could say that new opportunities lead to newer markets and newer uses for the same product category. ■

preethi_chamikutty@thinkdigit.com